

- The surface tension of a liquid decreases with increase in temperature and becomes zero at the critical temperature.
- On mixing soap, the surface tension of water decreases.

Some Phenomena Based on Surface Tension

- Warm soup is tasty because at high temperature its surface tension is low and consequently the soup spreads on all parts of the tongue.
- Medicines used for washing wounds, as dettol have a surface tension lower than water. So, they reach the fine cavities formed in the wound.
- The surface tension of the tooth-paste scum is also less that it spreads quickly on the full areas of teeth and clean them.
- Insects and mosquitoes swim on the surface of water in ponds and lakes due to surface tension.
- Hair of a shaving brush cling together due to surface tension, when the brush is taken out from the water.
- The ends of a glass tube become rounded on heating. Small drops of mercury are spherical while large ones are flat. Formation of lead shots, liquid-drop model of nucleus. Floatation of needle on water. Dancing of camphor on water and helping. Detergent in cleaning the clothes all are due to surface tension.

ANGLE OF CONTACT

The angle inside the liquid between the tangent to the solid surface and the tangent to the liquid surface at the point of contact is called the angle of contact for that pair of solid and liquid.

- The angle of contact for pure water and clean glass is zero. For ordinary water and glass it is about 8° .
- The angle of contact for water and silver is 90° .
- The liquids which wet the solid have acute angle of contact. Meniscus of these liquids will be concave.
- The liquids which do not wet the solid have obtuse angle of contact as for mercury and glass the angle of contact is 135° . Meniscus of these liquids will be convex.

CAPILLARITY

If a capillary tube is dipped in a liquid, liquid ascends or descends in the capillary tube. This phenomenon is called capillarity.

- The oil in the wick of a lamp rises due to capillary action of threads in the wick.
- To prevent loss of water due to capillary action, the soil is loosened and split into pieces by the farmers.
- The root hairs of plants draws water from the soil through capillary action.

- Writing nib is split in the middle, so that a fine capillary is formed in it. When it is dipped in ink, the ink rises in the capillary.

➤ **Note** The liquids which wet glass for which the angle of contact is acute, rise up in capillary tube, while those which do not wet glass, for which the angle of contact is obtuse are depressed down in the capillary.

VISCOSITY

It is the property of the liquid by virtue of which, it opposes the relative motion between its adjacent layers.

- Viscosity is the property of liquids and gases both.
- With rise in temperature, viscosity of liquids decreases and that for gases increases.
- Viscosity of liquid increases with increase in pressure.
- Viscosity of a fluid is measured by its coefficient of viscosity. It's SI unit is $(\text{N} \cdot \text{s} / \text{m}^{-2})$.

STOKES' LAW

When a small spherical body of radius r is moving slowly with a constant velocity v through a perfectly homogeneous medium (liquid or gas) of infinite extension and coefficient of viscosity η , then the retarding viscous force acting on the body is $F = 6\pi\eta rv$

- This law is used in Millikan's method for determining the electronic charge and for finding out the radii of small oil-drops by measuring their terminal velocity in air.

TERMINAL VELOCITY

When a body falls in viscous medium, its velocity first increases and finally becomes constant. This constant velocity is called terminal velocity. The terminal velocity of the sphere is directly proportional to the square of the radius of the sphere.

- The cloud particles fall down very slowly because of the viscosity of air and hence appear floating in the sky.
- Thin liquids like water, alcohol, etc are less viscous than thick liquids coal for blood, honey, glycerine, etc.
- Honey is less viscous than glycerine.

HEAT AND THERMODYNAMICS

TEMPERATURE

The measurement of hotness or coldness of a body is called its temperature. When two bodies are placed in contact, heat always flows from a body at higher temperature to the body at lower temperature.

THERMOMETER

An instrument used to measure the temperature of a body is called a thermometer. Mercury is generally used as, thermometric substance, because it is sensitive to slightest change in temperature which is notable when it is used in a thermometer. Mercury has a freezing point of -38.83°C and boiling point at 356.7°C , this gives a very big range to measure temperatures for many general purposes. Also, it does not stick to (wet) the capillary wall of the thermometers and it is easy to obtain in its elemental form.

Temperature Scales

Some temperature scales are as follows

1. **Celsius Scale (C)** The melting point of the ice at standard atmospheric pressure is regarded as 0°C and the boiling point of water as 100°C . This scale was designed by **Anders Celsius** in the year 1710.

The degree celsius ($^{\circ}\text{C}$) can refer to a specific temperature on the celsius scale as well as a unit to indicate a temperature interval.

2. **Fahrenheit Scale (F)** The melting point of ice is regarded as 32°F and the boiling point of water as 212°F . This scale was designed by **Gabriel Fahrenheit** in the year 1717.

3. **Reaumer Scale (R)** The melting point of ice is regarded as 0°R and the boiling point of water as 80°R . This scale was introduced by **RA Reaumer** in year 1730.

- Scales of temperature measurement,

$$\frac{C - 0}{100 - 0} = \frac{F - 32}{212 - 32} = \frac{R - 0}{80 - 0} = \frac{K - 273}{100 - 0}$$

$$\Rightarrow \frac{C}{5} = \frac{F - 32}{9} = \frac{R}{4} = \frac{K - 273}{5}$$

- The normal temperature of a human body is 37°C or 98.4°F .
- -40° is the temperature at which Celsius and Fahrenheit thermometers read same.
- The range of a laboratory thermometer is generally from -10°C to 110°C . A laboratory thermometer cannot be used to measure the human body temperature.

Total Radiation Pyrometer

Radiation pyrometer measures the temperature of a body by measuring the radiation emitted by the body. It cannot measure temperature below 800°C because at low temperature, emission of radiation is very small and cannot be detected. Sun's temperature is measured by it.

- **Note** Bolometer and thermopile are also used to measure the amount of heat radiated by a body.

HEAT

It is a form of energy which produces the sensation of warmth in our body.

- It is due to the kinetic energy of the molecules constituting the object or body. Its units are calorie, kilocalorie or joule.
- Calorie is the amount of heat required to raise the temperature of 1 g of water through 1°C (from 10°C to 11°C). It is represented by cal, $1 \text{ cal} = 4.18 \text{ J}$.
- Air is a non-conductor of heat and silver is the best conductor of heat.
- Ebonite rubber, glass and air are bad conductors of heat.

Specific Heat

- Specific heat is the amount of heat required to raise the temperature of a unit mass of the substance by 1°C .
- If Q heat changes the temperature of m by ΔT , then specific heat

$$C = \frac{Q}{m\Delta T}$$

The SI unit of specific heat is J/kg-K .

- Specific heat of gold = 130 J/kg-K
- Specific heat of water = 4180 J/kg-K
- The specific heat of water is maximum.
- Mercury has low specific heat.

Thermal Capacity

It is the amount of heat required to raise the temperature of a body through 1K . Its units are J/K or $\text{cal}/^{\circ}\text{C}$ or $\text{kcal}/^{\circ}\text{C}$.

Thermal capacity = mC

where, m is the mass of the body and C is the specific heat.

Water Equivalent

It is the mass of water, which absorbs or emits the same amount of heat as is done by the given body for the same rise or fall in temperature.

- The water equivalent of a body is denoted by W . Its SI unit is kg and CGS unit is g.
- Water equivalent, $W = mC$
where, m is mass of the body and C is the specific heat.

Principle of Calorimetry

The heat lost by the hot body must be equal to the heat gained by the cold body. This is known as principle of calorimetry or principle of mixtures, i.e.

$$\text{Heat gained} = \text{Heat lost}$$

- Work and heat are two equivalent forms of energy.
- Calorimeter is a cylindrical vessel made of copper.

THERMAL EXPANSION

The phenomenon of change in dimensions of an object due to heat supplied is known as thermal expansion.

- A solid can undergo three types of expansions:
 - Linear expansion (expansion in length)
 - Superficial expansion (expansion in area)
 - Cubical expansion (expansion in volume)
- Relation between the coefficients of linear, superficial and cubical expansion : $\alpha : \beta : \gamma = 1 : 2 : 3$

Practical Applications of Thermal Expansion

- When hot tea is poured into a glass tumbler made of soft glass, it cracks because of thermal expansion of glass.
- Telegraph wires are given enough gap to allow the wires for contraction in winter.
- An ordinary pendulum clock runs faster in winter but slower in summer, because in summer the length of pendulum increases, while in winter it decreases.
- A metallic ball before heating can pass easily through the ring, but after heating the ball does not pass through the ring because of the thermal expansion of ball.
- A gap is provided between the iron rails of the railway track, so that rails can easily expand during summer and do not bend. In the construction of bridges, ends of steel girders are not fixed but placed on rollers to allow free expansion and contraction in summer and winter respectively to avoid any damage to the bridge.
- A tightly fitted cork can be removed from a glass bottle without breaking it on heating its mouth carefully. The glass wall expands and the cork will come out easily.

Expansion of Liquids

- Liquids do not have their definite length and surface area. It always need a container. When they are heated, they expand and at the same time the container also expands.
- The expansion of a liquid, when the expansion of its container has not been taken into account is called **apparent-expansion**.
- The expansion of a liquid, when the expansion of its container is taken into account, is called **real expansion**.

Anomalous Expansion of Water

Almost every liquid expands with the increase in temperature. But when temperature of water is increased from 0°C to 4°C its volume decreases. If the temperature is increased above 4°C its volume starts increasing, clearly density of water is maximum at 4°C .

Interconversion of States of Matter

The temperature at which solid starts to liquify is known as melting point of that solid.

- The process in which a solid changes into a liquid on heating is called melting solid. Change of state takes place i.e. ice changes into water.
- The process in which a liquid changes into a solid is called **fusion**. The temperature at which liquid starts to freeze is known as **freezing point** of the liquid.
- The process in which a liquid changes into a vapour on heating is called **vaporisation**. The temperature at which the liquid starts to evaporate is called the **boiling point** where change of state takes place i.e. water changes into water vapour or steam.
- The process of change of state directly from solid to vapour (or gas) is known as **sublimation**.

LATENT HEAT

It is the amount of heat absorbed by a unit mass of the substance to change its state without change in temperature.

- SI unit of latent heat is joule per kilogram.
- Latent heat of fusion is 80 cal/g and vaporisation is 540 cal/g .
- Ice at 0°C is more effective in cooling a substance than water at 0°C .
- If Q is quantity of heat absorbed (or given out), then

$$Q = mL$$
 where, m = mass of the substance
 L = latent heat of substance.
- Steam contains more heat, in the form of latent heat, than boiling water.
- The burns caused by steam are much more severe than those caused by boiling water though both of them are at the same temperature of 100°C .

EVAPORATION

It is the slow process of conversion of a liquid into its vapour even below its boiling temperature.

- Cooking is caused by evaporation due to expansion.
- Factors affecting the evaporation of a liquid are
 - the area of the liquid surface
 - the nature of the liquid
 - temperature of the liquid
 - dryness of air
 - the movement of the air
 - temperature of air or surrounding

RELATIVE HUMIDITY

The ratio of amount of water vapour (m) actually present in a certain volume of air at a given temperature to the amount of water vapour (M) required to saturate it, is called relative humidity, R_H .

$$R_H = \frac{m}{M} \times 100\%.$$

- The amount of water vapour in air is called as **humidity**.
- The amount of water vapour in the air varies as it depends on the rate of evaporation.
- The amount of water vapour present in 1 m^3 air is called its **absolute humidity**.
- Relative humidity is measured by **hygrometer**.
- Relative humidity increases with the increases of temperature.

THERMODYNAMICS

Thermodynamics is a branch of physics which deals with exchange of heat energy between bodies and conversion of the heat energy into mechanical energy and *vice-versa*.

First Law of Thermodynamics The amount of heat given to a system is used up in two ways, first to increase the internal energy and second to do the external work. i.e.

$$dQ = dU + dW.$$

Second Law of Thermodynamics The second law of thermodynamics is the outcome of human experience under which heat energy can be converted into mechanical energy. This law is based upon the two statements given below.

- It is impossible to construct a device which operates in a cycle that will take heat from a body and convert it completely into the work. (Kelvin-Planck's Statements)
- It is impossible to construct a self acting device which operates in a cycle that will transfer heat from a cold body to a hot body without expenditure of work (Clausius Statement).

Nature of Internal Energy

- According to the kinetic model of matter, every substance is composed of molecules which are in constant motion inside the substance. Hence, molecules possess kinetic energy.
- Molecules also exert force on one another which is called intermolecular force, hence they possess potential energy also. The kinetic energy and potential energy of molecules is the internal energy of the substance.
- The internal energy of an ideal gas is only the kinetic energy of its molecules which depends only upon the temperature of the gas.

Difference between Petrol Engine and Diesel Engine

Petrol engine	Diesel engine
It works with a spark plug.	It works with an oil plug.
Efficiency is smaller (47%).	Efficiency is larger (55%).
It is associated with the risk of explosion, because petrol vapour and air is compressed. So, low compression ratio is kept.	No risk of explosion, because only air is compressed. Hence, compression ratio is kept large.
Petrol vapour and air is created with spark plug.	Spray of diesel is obtained through the jet.

Transmission of Heat

Transfer of heat from one place to other place is called transmission of heat.

There are three processes by which transmission of heat takes place.

(a) Conduction (b) Convection (c) Radiation

- In solids, transmission of heat takes place by **conduction process**.
- In liquids and gases heat takes place by **convection process**.
- Heat from the sun reaches on the earth by **radiation**.
- Mercury though a liquid is heated by conduction and not by convection.
- In rooms, ventilators are provided to escape the hot air by the process of convection.
- Air is poor conductor of heat.
- On a cold night two thin blankets gives more hotness than a single thick blanket because the layer of air between the two blankets works as better insulator of heat.
- Cooking utensils are made up of aluminium, brass and steel, because these substances have low specific heat and high conductivity.

KIRCHHOFF'S LAW

Kirchhoff's law signifies that good absorbers are good emitters. If a shining metal ball with some black spot on its surface is heated to a high temperature and seen in dark, the shining ball becomes dull but the black spot shines brilliantly because black spot absorbs radiation during heating and emit in dark.

- White and light colours are bad absorbers and good reflectors of heat.
- In deserts, day temperature is very high and night temperature is extremely low because the specific heat of sand is very low. Therefore, it absorbs the heat readily and its temperature raises by a large degree during day. At night sand radiates the heat equally readily making the temperature loss.

Perfectly Black Body A perfectly black body is one which absorbs completely all the radiations falling on its surface, whatever be the wavelength.

Perfectly black body is a perfect absorber hence, according to Kirchhoff's law, it will also be a perfect radiator. It is not essential that a perfectly black body should appear black. Sun emits radiation of all wavelength, so it may be called a black body, even though it looks white.

Stefan's Law The radiant energy emitted by unit area of perfectly black body per unit time is directly proportional to the fourth power of its absolute temperature.

$$E \propto T^4 \Rightarrow E = \sigma T^4$$

where, σ is Stefan's constant.

- **Note** Wein's displacement law can be used to compute the temperature of the sun or of the stars.

Newton's Law of Cooling The rate of loss of heat by a body is directly proportional to the difference in temperature between the body and the surrounding.

If the door of refrigerator is kept open, it will not cool the room, it will increase the temperature of the room, because heat rejected by the refrigerator to the room, will be more than the heat taken by the refrigerator from the room.

OSCILLATIONS AND WAVES

PERIODIC MOTION

When a body repeats its motion continuously on a definite path in a definite interval of time, then its motion is called **periodic motion** and the interval of time is called **time period**. Motion of hands of a clock, motion of earth around the sun and motion of the needle of a sewing machine are examples of periodic motion.

Oscillatory Motion

If a body in periodic motion moves along the same path to and fro about a definite point (mean position or equilibrium position), then the motion of the body is a vibratory motion or oscillatory motion. Oscillatory motion is also called as harmonic motion.

- The motion of the pendulum of a wall clock, the motion of the bob of a simple pendulum, the motion of a loaded spring, the motion of liquid contained in a U-tube, the motion of the prongs of tuning fork and the motion of a bar magnet suspended in the earth's magnetic field are examples of oscillatory motion.
- All oscillatory motions are periodic motions, but all periodic motions are not oscillatory.

SIMPLE HARMONIC MOTION (SHM)

If a particle repeats its motion about a fixed point after a regular time interval in such a way that at any moment the acceleration of the particle is directly proportional to its displacement from the fixed point at that moment and is always directed towards the fixed point, then the motion of the particle is called simple harmonic motion.

- Vibrations of the prongs of a tuning fork, oscillations of a body. Suspended by a spring. Oscillations of a body partly immersed in a liquid and oscillations of a simple pendulum are examples of simple harmonic motion.
- When a particle executing SHM passes through the mean position, then

- no force acts on the particle.
- velocity is maximum.
- acceleration is zero.
- kinetic energy is maximum.
- potential energy is zero.

- When a particle executing SHM is at the extreme end, then
 - velocity of particle is zero.
 - acceleration of the particle is maximum.
 - kinetic energy of particle is zero.
 - potential energy is maximum.
 - restoring force acting on particle is maximum.
- In case of spring block system, time period of oscillation is given by $T = 2\pi \sqrt{\frac{m}{k}}$, where m is the mass of the block.
- In case of spring block system, the restoring force $F = -kx$ where, x is displacement of the block from mean position and k is spring constant.

RESTORING FORCE

When oscillating particle is displaced from its equilibrium position, then a periodic force acts upon it which is always directed towards the equilibrium position. This force is called **restoring force**.

Simple Pendulum

A simple pendulum consists of a small metal ball suspended by a long thread from a rigid support, such that the bob is free to swing back and forth.

- The distance of the pendulum from its mean position is called its **displacement**.
- The maximum displacement of the pendulum on either side of its mean position is called the **amplitude** of oscillations.
- **Time period** is the time taken by the pendulum to complete one full oscillation.